

Computational Control Certificate

CTRL-02 — Complete-split cover value

Erdős #81, Paper II — internal validation evidence

Reference. Paper II, Proposition 6.2 and Corollary 6.3

Date. 2026-07-07 **Verdict.** **PASS**

Statement audited

$\tau_3^*(S_{p,q}) = \frac{P+pq}{3}$ for $q \leq p-1$, $= P$ for $q \geq p-1$ (with $P = \binom{p}{2}$), and $= 0$ in the triangle-free corners $p \leq 1$ and $(p,q) = (2,0)$; and the induced values of $\Phi_\tau(S_{p,q})$.

Method

For a grid of (p,q) the value $\tau_3^*(S_{p,q})$ is computed by exact linear programming and compared against the closed form; $\Phi_\tau = |E| - 2\tau_3^*$ is compared against Corollary 6.3. The degenerate case $S_{2,0} = K_2$ is checked explicitly.

Verbatim output

p	q	LP tau3*	formula	Phi_LP	Phi_formula	
0	1	0.0000	0.0000	0.000	0.000	OK
0	2	0.0000	0.0000	0.000	0.000	OK
0	3	0.0000	0.0000	0.000	0.000	OK
0	4	0.0000	0.0000	0.000	0.000	OK
1	0	0.0000	0.0000	0.000	0.000	OK
1	1	0.0000	0.0000	1.000	1.000	OK
1	2	0.0000	0.0000	2.000	2.000	OK
1	3	0.0000	0.0000	3.000	3.000	OK
1	4	0.0000	0.0000	4.000	4.000	OK
2	0	0.0000	0.0000	1.000	1.000	OK
2	1	1.0000	1.0000	1.000	1.000	OK
2	2	1.0000	1.0000	3.000	3.000	OK
2	3	1.0000	1.0000	5.000	5.000	OK
2	4	1.0000	1.0000	7.000	7.000	OK
3	0	1.0000	1.0000	1.000	1.000	OK
3	1	2.0000	2.0000	2.000	2.000	OK
3	2	3.0000	3.0000	3.000	3.000	OK
3	3	3.0000	3.0000	6.000	6.000	OK
3	4	3.0000	3.0000	9.000	9.000	OK
checked (p,q) pairs				: 72		
tau3* LP vs formula mismatches:				0		
Phi_tau LP vs formula mismatch:				0		
degenerate S_2,0 (K_2): LP tau3*=0.0000 (formula 0)						
RESULT:				PASS		

Reproducibility

The exact script `control.py` and its verbatim output `results.txt` are included in this folder. Re-running `python3 control.py` reproduces `results.txt`. This certificate reports a computational

check on finite instances; it is supporting evidence, not a substitute for the analytic proof in Paper II. File integrity is recorded in `SHA256SUMS.txt` (this folder) and in the master `SHA256SUMS.txt` of the package.